

STOCHASTIC METHODS IN WATER RESOURCES

Unit 3: Random functions

Lecture 10: Spectral representation of random functions

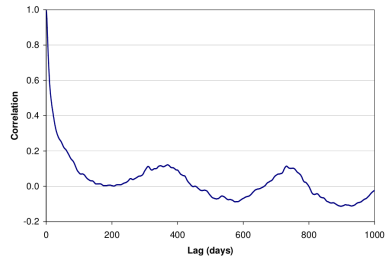
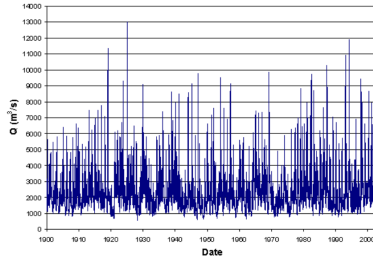
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Spectral representation of random functions: Definitions

- ▶ **Correlation functions** of a **time series**, when analyzed for longer period exhibit a **periodic behaviour** e.g. **Hole effect model** (see tables in lecture 9).
- ▶ For instance, the figures below show a time series of daily average streamflows and its **correlation function**.



- ▶ A clear periodic behaviour is observed in the **correlation function** which is also apparent in the time series.
- ▶ Such periodicity is caused because evaporation, which is driven by radiation and temperature, has a clear seasonal character, specially, at higher and lower latitudes and temperate climates.
- ▶ In conclusion, most hydrological time series show a **seasonal variation**. This means that to analyse these models with **stationary random functions** requires that this **seasonality is removed**.
- ▶ The occurrence of **seasonality** has also triggered the use of **spectral methods** in stochastic modelling, although it must be stressed that spectral methods are also very suitable for analysing stationary random functions.

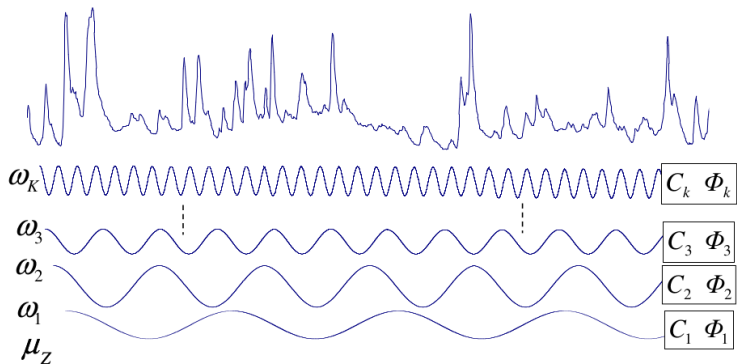
Spectral density function

The **spectral representation** of a **stationary random function** $Z(t)$, means that $Z(t)$ is composed by the sum of its mean μ_Z and $2K$ **sine waves** with **increasing frequencies**, where each frequency has a **random amplitude** C_K and a **random phase angle** Φ_K . So $Z(t)$ is expressed as

$$Z(t) = \mu_Z + \sum_{k=-K}^K C_K \cos(\omega_K t + \Phi_K)$$

where $C_K = C_{-K}$, $\Phi_K = \Phi_{-K}$ and $\omega_K = \pm \frac{1}{2}(2k - 1)\Delta\omega$.

The figure below is a schematic representation of the spectral decomposition of a stationary random function. The random signal is decomposed into its harmonics of increasing frequency (ω_K) with random amplitude C_K and random phase angle Φ_K .



$$Z(t) = \mu_Z \pm C_1 \cos(\omega t + \Phi_1) \pm C_2 \cos(2\omega t + \Phi_2) \pm \dots \pm C_K \cos(K\omega t + \Phi_K)$$

Spectral density function

Based on the mathematical representation of the spectral decomposition of the random process, the following relationships known as the **Winer-Khinchine relations** are obtained

$$C_Z(\tau) = \int_{-\infty}^{\infty} S_Z(\omega) \cos(\omega\tau) d\omega$$
$$S_Z(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} C_Z(\tau) \cos(\omega\tau) d\tau$$

where the function $S_Z(\omega)$ is known as the **spectral density function** of the **random process**. Note that the equations above show that **covariance function** C_Z is a **Fourier transform** of the **spectrum** and viceversa. These two equations form a **Fourier pair**. Making the lag $\tau = 0$, it is possible to understand the physical meaning of $S_Z(\omega)$ because

$$C_Z(0) = \int_{-\infty}^{\infty} S_Z(\omega) d\omega$$

from where it can be seen that the **variance** of the random function is equal to the integral over the **spectral density**. This means that the **variance** is a **weighted sum** of variance components, where each component consists of a **random harmonic function** of a given **frequency**. The **spectral density** then represents the weight of each of the attributing **random harmonics**, i.e. the **relative importance** of each **random harmonic** in **explaining** the **total variance** of the **random function**.

Spectral density function

It is easy to see the analogy with the **electromagnetic spectrum** where the **total energy** of **electromagnetic radiation** (which is analogous to the variance of our random signal) is found by the **area under the spectrum** and can be attributed to **relative contributions** from different **wavelengths**.

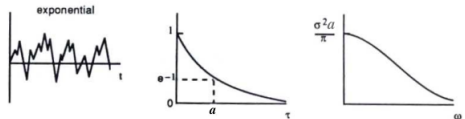
The following table shows expressions for **spectral density functions** for different **covariance functions**.

Exponential covariance	$C_Z(\tau) = \sigma_Z^2 e^{- \tau /a}$ for $a > 0$	$S_Z(\omega) = \frac{a\sigma_Z^2}{\pi(1+a^2\omega^2)}$
Random harmonic: $Z(t) = a \cos(\omega_0 t + \phi)$ where a and ω_0 are deterministic constants and ϕ is random.	$C_Z(\tau) = \frac{a^2}{2} \cos(\omega_0 \tau)$ for $\tau \geq 0$ and $a, \omega_0 > 0$	$S_Z(\omega) = \frac{a^2}{4} \delta(\omega - \omega_0)$
Hole effect (wave) model	$C_Z(\tau) = \sigma_Z^2 (1 - \tau /a) e^{- \tau /a}$ for $a > 0$	$S_Z(\omega) = \frac{a^3 \sigma_Z^2 \omega^2}{\pi(1+a^2\omega^2)^2}$
White noise model	$\rho_Z(\tau) = \begin{cases} 1, & \text{if } \tau = 0 \\ 0, & \text{if } \tau > 0 \end{cases}$	$S_Z(\omega) = c$

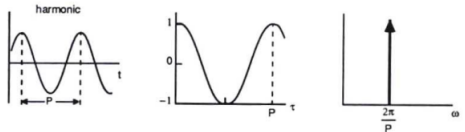
Spectral density function

The following figure shows typical realisations of the random functions involved, their correlation function and the associated spectrum.

- ▶ Note that the **spectrum** of **white noise** is a **horizontal line**, implying an infinite variance according to the equation.



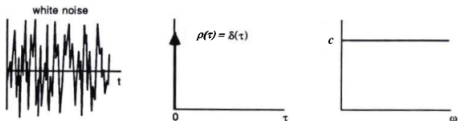
- ▶ This shows that **white noise** is a **mathematical construct**, and not a feasible physical process: the **area under the spectrum** is a measure for the **total energy of a process**.



- ▶ This area is infinitely large, such that all the energy in the universe would not be sufficient to generate such a process.



- ▶ In practice one often talks about **wide band processes**, where the **spectrum** has a **wide band of frequencies**, but encloses a **finite area**.



Spectral density function

Note that **spectral density function** is an even function as the **covariance function**: $S_Z(\omega) = S_Z(-\omega)$. This means that the **spectral density function** can be expressed as a **one-sided function** $G_Z(\omega) = 2S_Z(\omega)$, for $\omega \geq 0$. The **Wiener-Khinchine relations** become

$$C_Z(\tau) = \int_0^{\infty} G_Z(\omega) \cos(\omega\tau) d\omega$$
$$G_Z(\omega) = \frac{2}{\pi} \int_0^{\infty} C_Z(\tau) \cos(\omega\tau) d\tau$$

Sometimes it is convenient to work with the **normalised spectral density functions**, by dividing the **spectra** by the **variance**: $s_Z(\omega) = \frac{S_Z(\omega)}{\sigma_Z^2}$ and $g_Z(\omega) = \frac{G_Z(\omega)}{\sigma_Z^2}$. There is a relation between the **normalised spectral density function** $s_Z(\omega)$ and the scale of fluctuation. So if $\omega = 0$ we obtain

$$s_z(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \rho_Z(\tau) d\omega = \frac{\theta}{2\pi}$$

Formal spectral representation

A more formal representation of the **spectral density** is used in the literature based on **complex calculus**. Here the random function $Z(t)$ is defined as the **real part** $Re.$ of a **complex random function** $Z^*(t)$.

$$Z(t) = ReZ^*(t) = Re \left\{ \mu_Z + \sum_{k=-K}^K X_k e^{i\omega_k t} \right\} \xrightarrow{K \rightarrow \infty} Re \left\{ \mu_Z + \int_{-\infty}^{\infty} e^{i\omega t} dX(\omega) \right\}$$

where $\omega_k = \frac{1}{2}(2k - 1)\Delta\omega$ is the frequency, and X_k a **complex random number** representing the **amplitude**. This equation means that the **complex random process** is decomposed into a large number of complex harmonic functions $e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$ with **random complex amplitude**. According to this representation, the **Wiener-Khinchine** equations are

$$C_Z(\tau) = \int_{-\infty}^{\infty} S_Z(\omega) e^{i\omega\tau} d\omega$$
$$S_Z(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} C_Z(\tau) e^{i\omega\tau} d\tau$$

It can be demonstrated that these equations are equivalent to the prior **Wiener-Khinchine** the equations.

Estimating the spectral density function

For **stationary random functions**, the **spectral density function** can be estimated from the **covariance function** as

$$S_Z(\omega_i) = \frac{1}{2\pi} \left(\lambda_0 C_Z(0) + 2 \sum_{k=1}^M \lambda_k \hat{C}_Z(k\Delta\tau) \cos(\omega_i k) \right)$$

where $\omega_i = i\pi/M$ and $|i| = 1, 2, \dots, M$. The weights λ_k are necessary to smooth the covariances before performing the transformation. So that, a smooth **spectral density function** is obtained displaying only the relevant features. There are numerous ways to estimate the **smoothing weights**, the most common are:

► **Tukey window**

$$\lambda_k = \frac{1}{2} \left(1 + \cos \left(\frac{k\pi}{M} \right) \right), \quad k = 1, 2, \dots, M$$

► **Parzen window**

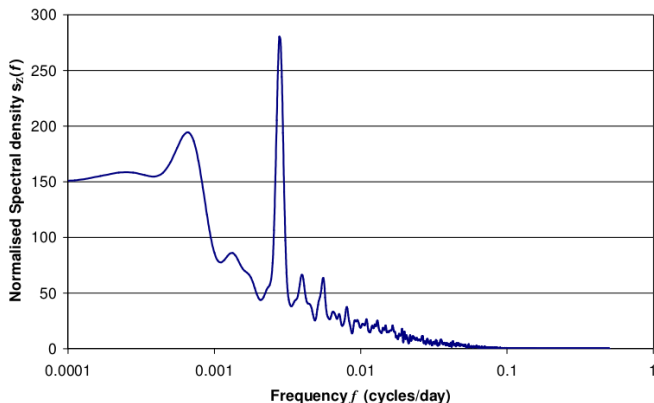
$$\lambda_k = \begin{cases} 1 - 6 \left(\frac{k}{M} \right)^2 + 6 \left(\frac{k}{M} \right)^3, & 0 \leq k \leq M/2 \\ \left(2 - \frac{k}{m} \right)^3, & M/2 \leq k \leq M \end{cases}$$

The **highest frequency** that is analysed is equal to $f_{\max} = \frac{\omega_{\max}}{2\pi} = 0.5$, which is the **highest frequency** that can be estimated from a time series: half of the frequency of the observations. This frequency is called the **Nyquist frequency**. For instance, if streamflow is observed once per day, the highest frequency that can be detected is on cycle per two days.

Estimating the spectral density function

The **smallest frequency** (largest wavelength) that can be analysed depends on the discretisation $f_{min} = \pi/(2\pi M) = 1/2M$, where M is the cutoff level (maximum lag considered) of the covariance function. The width of the **smoothing windows** (smoothing weight) is adjusted accordingly.

The figure below shows the spectrum of a daily discharge time series. The **spectral density function** was estimated using a **Parzen window** with $M = 9000$. Clearly, small frequencies dominate with a small peak between 4 and 5 years. Most prominent of course, as expected, there is a peak at a frequency of once a year, which exemplifies the strong seasonality in the time series.



Spectral representations of random space functions

Extending the equations above to two dimensions, the **stationary random function** $Z(x_1, x_2)$ can be expressed in terms of random harmonics as

$$Z(x_1, x_2) = \mu_Z + \sum_{k_1=-K_1}^{k_1=K_1} \sum_{k_2=-K_2}^{k_2=K_2} C_{12} \cos(\omega_{k_1} x_1 + \omega_{k_2} x_2 + \Phi_{12})$$

where C_{12} and Φ_{12} are the **random amplitude** and the **phase angle**, respectively, belonging to frequency ω_{k_i} , $i = 1, 2$

$$\omega_{k_i} = \pm \frac{2k_i \Delta \omega_i - 1}{2}$$

The **Wiener-Khinchine** equations then become

$$C_Z(h_1, h_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S_Z(\omega_1, \omega_2) \cos(\omega_1 h_1 + \omega_2 h_2) d\omega_1 d\omega_2$$
$$S_Z(\omega_1, \omega_2) = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} C_Z(h_1, h_2) \cos(\omega_1 h_1 + \omega_2 h_2) d\omega_1 d\omega_2$$

The **variance** is given by the volume under the **2D spectral density function** $S_Z(\omega_1, \omega_2)$. Therefore

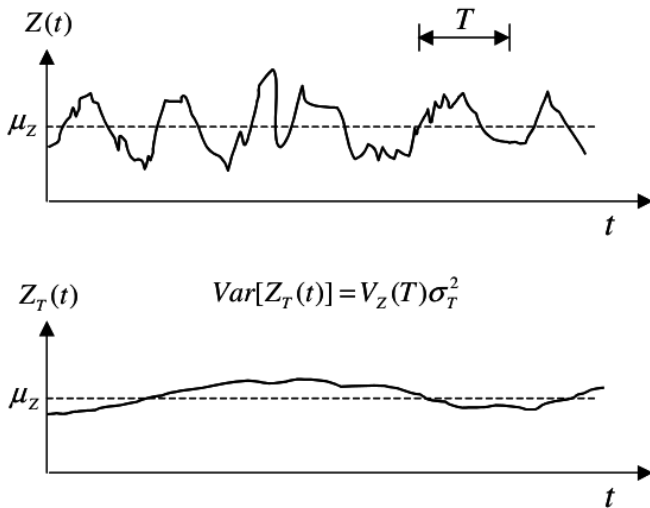
$$\sigma_Z^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S_Z(\omega_1, \omega_2) d\omega_1 d\omega_2$$

Local averaging of stationary random functions

For a stationary random function $Z(t)$, the local moving averaging function $Z_T(t)$ of $Z(t)$ is defined as

$$Z_T(t) = \frac{1}{T} \int_{t-T/2}^{t+T/2} Z(\tau) d\tau$$

The figure below shows the Local (moving) averaging of a stationary random function.



Local averaging of stationary random functions

Note that **local averaging** of a **stationary random process** will not affect the **mean**, but it does reduce the **variance**. So the **variance** of the **averaged process**

$\text{Var}[Z_T(t)] = E[(Z_T(t) - \mu_Z)^2]$ can be calculated for $\mu_Z = 0$ as

$$\begin{aligned}\text{Var}[Z_T(t)] &= E \left[\frac{1}{T} \int_{t-T/2}^{t+T/2} Z(\tau_1) d\tau_1 \frac{1}{T} \int_{t-T/2}^{t+T/2} Z(\tau_2) d\tau_2 \right] \\&= E \left[\frac{1}{T} \int_0^T Z(\tau_1) d\tau_1 \frac{1}{T} \int_0^T Z(\tau_2) d\tau_2 \right] \\&= \frac{1}{T^2} \int_0^T \int_0^T E[Z(\tau_1)Z(\tau_2)] d\tau_1 d\tau_2 \\&= \frac{1}{T^2} \int_0^T \int_0^T C_Z(\tau_2 - \tau_1) d\tau_1 d\tau_2\end{aligned}$$

From $\text{Var}[Z_T(t)]$ a new function is introduced called the **variance function**

$$V_Z(T) = \frac{\text{Var}[Z_T(t)]}{\sigma_Z^2}$$

The **variance function** thus describes the reduction in variance when averaging a random function as a function of the **averaging interval T**.

Local averaging of stationary random functions

The **variance function** is related to the **correlation function** as

$$V_Z(t) = \frac{1}{T^2} \int_0^T \int_0^T \rho_Z(\tau_2 - \tau_1) d\tau_1 d\tau_2$$

This equation can be simplified as

$$V_Z(t) = \frac{1}{T} \int_0^T \left(1 - \frac{\tau}{T}\right) \rho_Z(\tau) d\tau$$

The table below gives a number of **correlation functions** and their **variance functions**.

Exponential (first order autoregressive process)	$\rho_Z(\tau) = e^{-\tau/a}$ for $\tau \geq 0$, $a > 0$	$V(T) = 2 \left(\frac{a}{2}\right)^2 \left(\frac{T}{a} - 1 + e^{-T/a}\right)$ $\theta = 2a$
Second order autoregressive process	$\rho_Z(\tau) = \left[1 + \frac{\tau}{a}\right] e^{-\tau/a}$ for $\tau \geq 0$, $a > 0$	$V(T) = \frac{2a}{T} \left[2 + e^{-T/a} - \frac{2a}{T} (1 - e^{-T/a})\right]$ $\theta = 4a$
Gaussian correlation function	$\rho_Z(\tau) = e^{-\tau^2/a^2}$ for $\tau \geq 0$, $a > 0$	$V(T) = \left(\frac{a}{T}\right)^2 \left[\sqrt{\pi} \operatorname{Erf}\left(\frac{T}{a}\right) - 1 + e^{-T^2/a^2}\right]$ $\theta = a\sqrt{\pi}$

where $\tau = |t_1 - t_1|$.

Local averaging of stationary random functions

If we analysed the **variance function** when $T \rightarrow \infty$, $\lim_{T \rightarrow \infty} V_Z(T) \rightarrow \frac{\theta}{T}$, where θ is the **scale of fluctuation**. The **scale of fluctuation** is also given for the three correlation models given in the table above. The **scale of fluctuation** was already introduced earlier as a measure of **spatial correlation** of the **random function** and can also be calculated using the **correlation function** or the **spectral density**. $V_Z(T) \rightarrow \frac{\theta}{T}$ states that for larger averaging intervals the variance reduction through averaging is inversely proportional to the **scale of fluctuation**: the larger the **scale of fluctuation** the larger T should be to achieve a given variance reduction.